

Particles and Nuclei 2

Physics 7502

Instructor: Prof. Justin Frantz

Spring 2026

Particles and Nuclei 2

- First Day 1/13/26
- Announcements:
- Canvas – available
- Not finalized but could start thinking of those Griffiths problems
- INITIAL READING Griffiths Ch 1 by ~Thursday. Mann 1.2 and 1.3.
- Preview of some HW 1 problems available ~tonight if not whole set

Syllabus

- Website
 - we will use Canvas
 - Slides posted there regularly
 - Slides content exceeds textbooks
- Syllabus
- Other Syllabus-related things
 - 1st Reading Assignment Griff. Chapter 1 (For next class)
 - 1st HW Assignment posted later this week
 - Due ~1 Wk Later
 - Grading Percentages Not Completely Fixed
- Schedule pre-post Spring Break

Particles and Nuclei 2

- OU Nuclear grad courses (traditionally):
 - Particles and Nuclei 1 -- Nuclear Physics
 - Particles and Nuclei 2 -- Particle Physics
- However distinction btw two topics gets confusing
 - See first paragraph of Intro to my thesis (linked on Bb)
- Applied, and even any pure, particle physics that involves the strong nuclear force / QCD (defined soon) is often referred to also as “Nuclear Physics” (NP)
 - Most importantly, by the Department of Energy/NSF, ie the two main US funding agencies



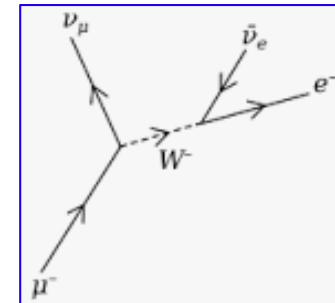
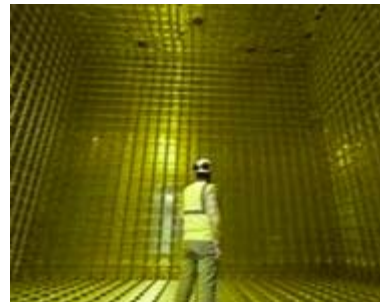
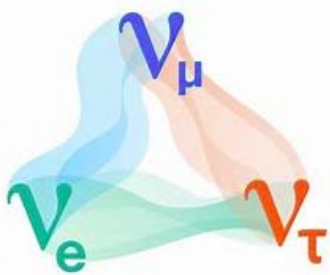
Distinction of Physics by US DOE Funding Divisions

- Nuclear vs Particle in US defined by Dept of Energy Funding divisions
 - High Energy vs Nuclear Physics
- Other countries are similar but have some different divisions for the various particle sub areas – e.g. neutrinos, etc...
- US still biggest SINGLE-SOURCE funder in the world – \$6-7 B
 - Canada – much smaller population 39 M = California or OH+PA+NY
 - Europe ? CERN budget 1.2 B
 - China trying get in game
- Interesting : NASA budget \$25.1 Billion!



Particle Physics – Energy (D.o.) (HEP)

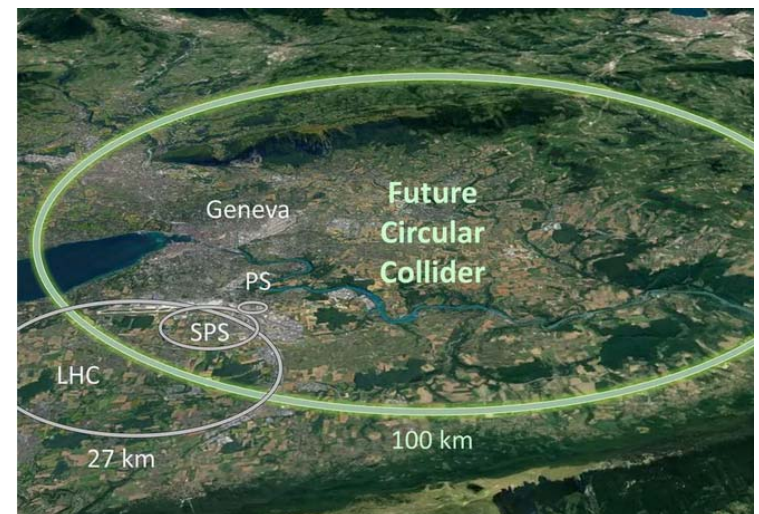
- Cleaner distinction: by Energy? These define communities
- High Energy Physics (HEP) → “pure” Particle
 - ~\$2–3 Billion DOE annual budget
 - <https://www.energy.gov/science/hep/high-energy-physics>
 - HEPAP – HEP Advisory Panel (NSF + DOE)
 - “P5” Particle Physics Project Prioritization Panel – Strategic Plan 2023
 - <https://www.usparticlephysics.org/>
- Things included:
 - LHC
 - Neutrino physics (largest?–ish now?) DUNE.
 - But DUNE was having problems budget is doubling from \$3 B to \$6 B – this is very similar to numbers that got the Superconducting Super- Collider (SSC) got cannnded by Congress
 - FUTURE → Higg’s Factories / 10 TeV pCM



HEP → Future / 10 TeV “pCM”

- Higg's factories Higgs discovered 2012 Mass 125 GeV
- Nominally Ebeam $125/2 = 62.5$ GeV → RHIC ?
 - But for protons like at LHC/RHIC quarks inside only carry small fraction of Ebeam.
- “pCM” : ‘parton’ CM energy → Future Circular Collider
~100 TeV proton CM
- Use Lepton: previous p5 high priority :
 - Long time: ILC – International Linear Collider e+/- no Bremstrahlung
 - LEP: (former LHC: 100 GeV e beams → circular doable – China!
- MUON COLLIDER!
 - Like radioactive nucl. beam/ FRIB
- Plasma Wake acceleration
- [COLLIDERS THOUGH!]
 - Relativity : “fixed target → $\sqrt{E_{\text{beam}}}$ ”

Recommendation 4: Invest in a comprehensive initiative to develop the resources—theoretical, computational, and technological—essential to realizing our 20-year strategic vision. This includes an aggressive R&D program that, while technologically challenging, could yield revolutionary accelerator designs that chart a realistic path to a 10 TeV pCM collider.



DOE Nuclear Physics (NP)

- Cleaner distinction: by $\sim$ Energy? These define communities
- Nuclear Physics (NP)
 - $\sim$ 2-3 Billion DOE annual budget
 - <https://www.energy.gov/science/np/nuclear-physics>
 - NSAC Nuclear Science Advisory Committee (NSF + DOE)
 - Long Range Plan (10 yr plan every 7 yrs)
 - (2023 – Nuclear Lunch we will cover this semester)
- “Medium Energy” (MEP) and “Low” Energy -- $>$ NP
- **EXCEPT: MEP** : 1) RHIC “HEP-Nuclear” 2) Jlab (some remnants of SLAC/HERA e+p)



(Most recognizable
RHIC logo)



DOE NP continued

- Current MEP Program areas under DOE NP: LE/ME/RHI
- Budget split 3 ways but majority alternates around (RHIC ~ 1 Billion/yr → ramping down, construction FRIB - ~3B, JLAB 12 GeV Upgrade - \$2 B)
- **New: Facility:** Electron Ion Collider!!!
 - will become it's own new program area
 - Expected to take most of/much of HE+ME
 - RHIC ending, Jlab slowing, US-LHC ?
 - Asymmetric beams: 10-18 GeV e ~100 GeV p (or per nucleon A)



Map of QCD Phenomena

Heavy Ions Motivation 1: Anything Else?

Yes!!!

“Direct” q/g
interactions

Hadron Structure

• Exotic QCD Matter !

- o Heavy Ion: QCD Plasma

- o Neutron Stars/ Superconductors

Decreasing Energy

Nuclear Structure/
Hadronic Bound States

Fragmentation/
Hadronization

All DOE “Nuclear” Phys (NP)

Break Exercise

- Test of tophat
- Read first paragraph of intro of my thesis on Blackboard
- On tophat enter...
 - ~3 word summary of 5th sentence
 - the 3rd to last word of that paragraph

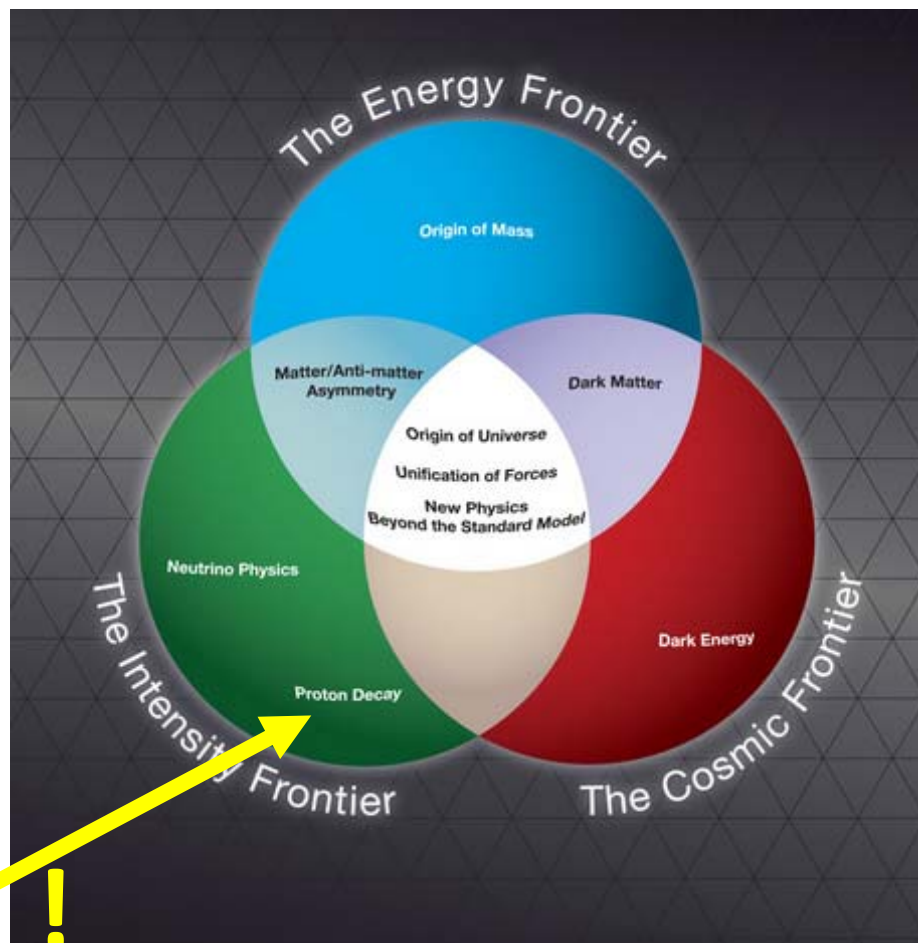
A couple other messages about the course (discussed in class):

-(see previous slides) Modern high and medium energy particle physics uses COLLIDERS (beam-beam) ~All textbooks available are written aimed towards older "fixed target" (single beam) situations.

-Although there are diff particles too – though these are pretty easy to understand– the new PHYSICS compared to low energy nuclear physics is ALL basically effects of special relativity, ie relativistic effects. Keep that in mind, much of the course is essentially a study of relativistic effects.

- the most important of those effects is also that there is generally enough energy to form new particles, thus the need for quantum field theory, also with its many relativistic complications.

More, visualizations



<https://science.osti.gov/hep/About/Vision-for-HEP>

Similar distinction NP/HEP EU?

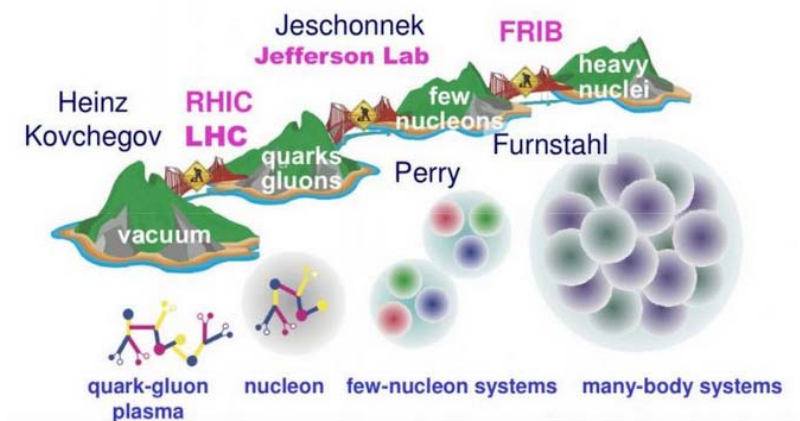
LHC - 1 out of 12 months NP/RHI

11/12 months

p+p/HE

Weird rules for HEP vs NP collaborator
“dues”

Particle and Nuclear Astrophysics
Dark Matter



OSU nuclear theory
page 2yrs ago

Particles and Nuclei 2 -Topics

- OU Nuclear grad courses (traditionally):
 - Particles and Nuclei 1 -- Nuclear Physics
 - Particles and Nuclei 2 -- Particle Physics
- In this course there will be a decided focus on the NP-side of Particle Physics
- **Topics:**
 - 2/3 Griffiths book material, will focus less on Weak Force chapter
 - QCD Jets
 - DIS Physics
 - Heavy Flavor & Heavy **Exotica***
 - Neutrino Physics
 - Non-perturbative/Lattice QCD
 - UHECR (HE cosmic ray)*
 - Dirac Equation Spinors
 - HE Hadronic collisions
 - Jlab and Spin Physics
 - Electron Ion Collider Physics
 - Relativistic Heavy Ion Physics
 - DetectorsAccelerators
 - Diffractive/Pomeron Particle Physics

*maybe

QFT: MAJOR COOL QFT THINGS